

Why is HTN
important?

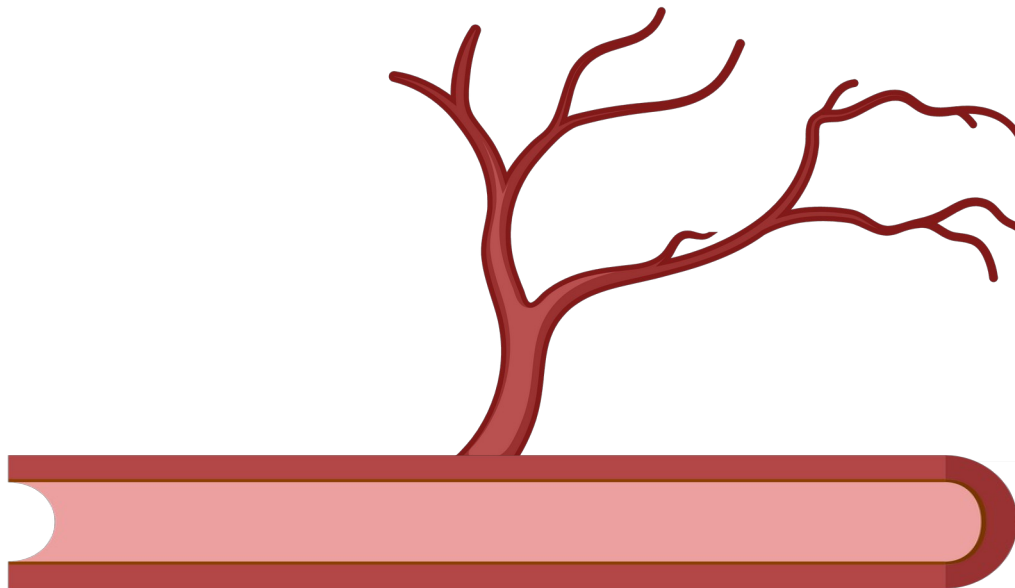
Treatment

Secondary causes

CASES

Hypertension

Sustained **BP** >130/80 (ACC/AHA)



Why is HTN important?

Treatment

Secondary causes

CASES

What are the benefits of treatment?

Example: 50-year-old woman who smokes tobacco, otherwise healthy with sustained BP >150/90

→ If treated to BP <140/85

What are the consequences of sustained HTN?

What percent of the population has HTN?

45%
of Americans >18y

CV Events Over 10 years

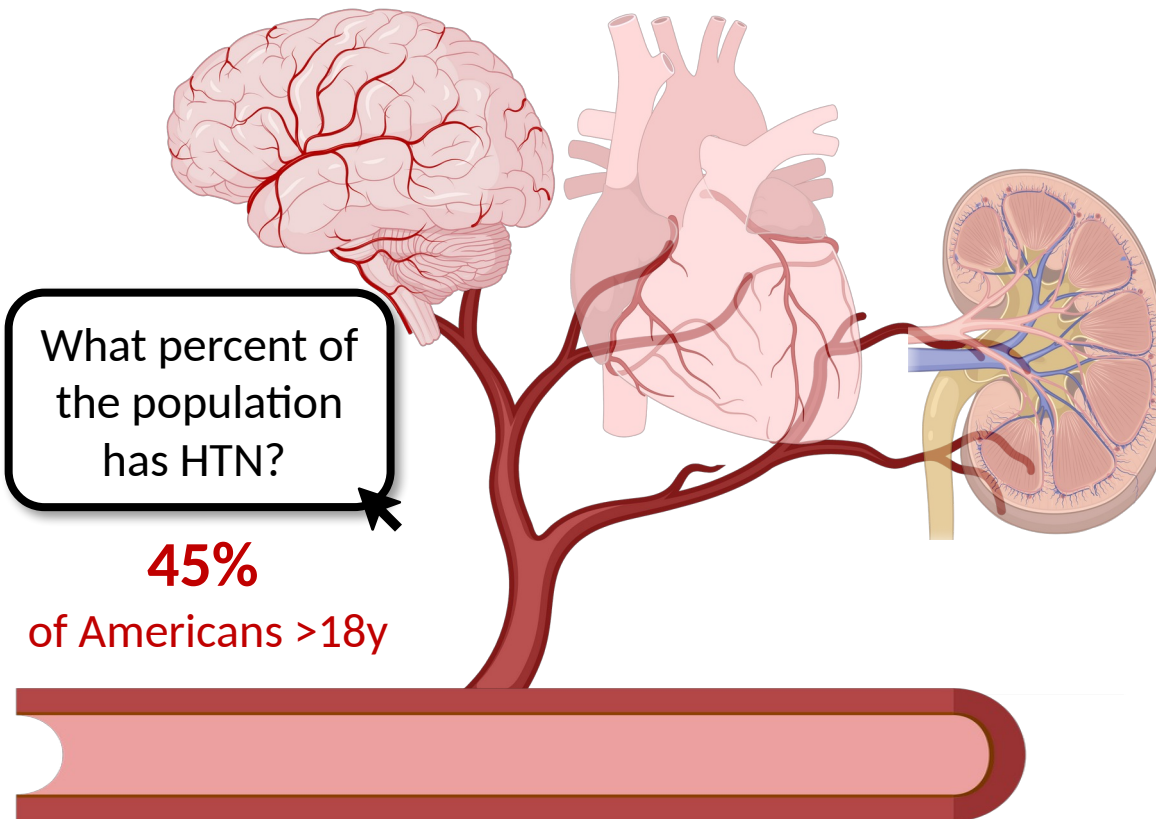
Absolute risk reduction = 10%

NNT = 10

Mortality Over 10 years

Absolute risk reduction = 6%

NNT = 16



Treatment Target: Based on BP & Risk

Ideal blood pressure (if achieved through lifestyle alone)?

$\text{SBP} < 120$
 $\text{DBP} < 80$

...but using medications to achieve that target is not necessarily worth the risks. What are the **pharmacologic treatment targets for:**

Patients without
high-risk diagnoses

$\text{SBP} < 140$
 $\text{DBP} < 90$

High-risk
patients?

$\text{SBP} < 130$
 $\text{DBP} < 80$

Who is considered
"high risk"?

Diabetes

Prior CVA
or MI

CKD

Heart
Failure

10y ASCVD
risk >10%

Why is HTN
important?

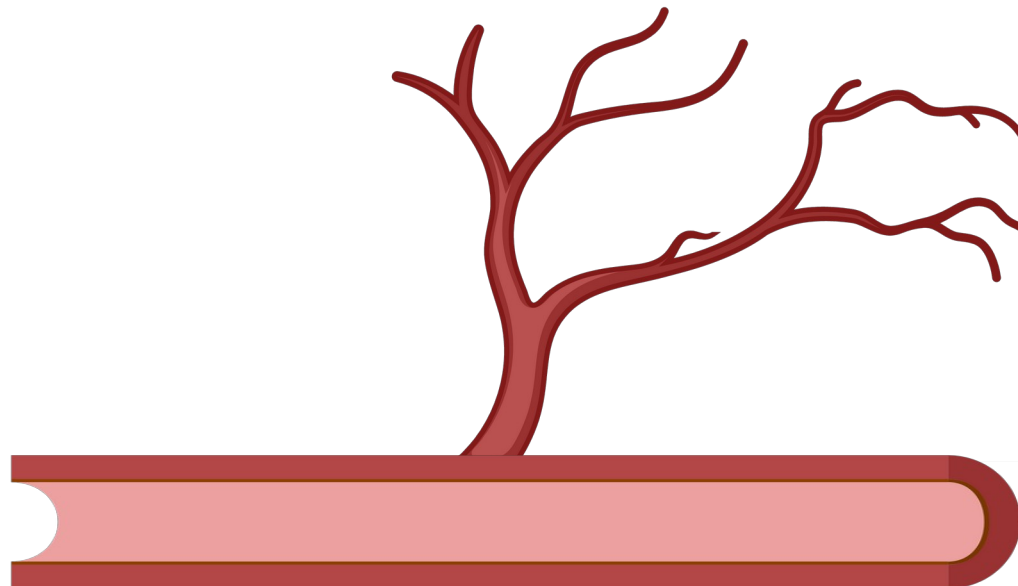
Treatment

Goals

1st line agents

Secondary causes

CASES



Why is HTN important?

Treatment

Goals

1st line agents

Dihydro CCB

ACE-I

ARB

Thiazide

Secondary causes

CASES

Four First Line Agents

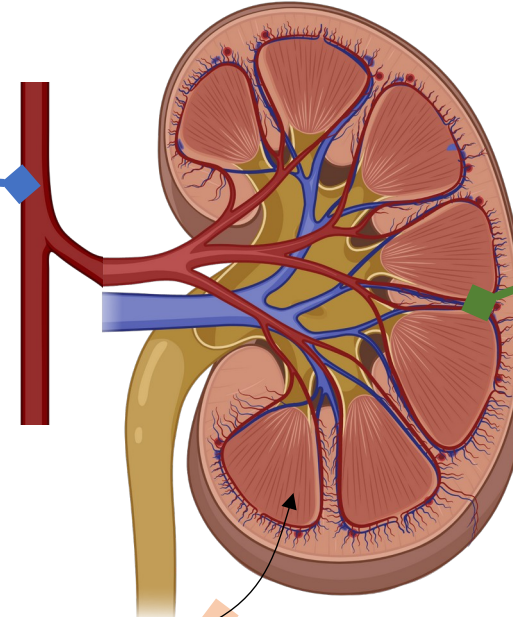
Calcium Channel Blocker (CCB)

Thiazide

ACE Inhibitor

Angiotensin Receptor Blocker (ARB)

Angiotensin II



Why is HTN important?

Treatment

Goals

1st line agents

Dihydro CCB

ACE-I

ARB

Thiazide

Secondary causes

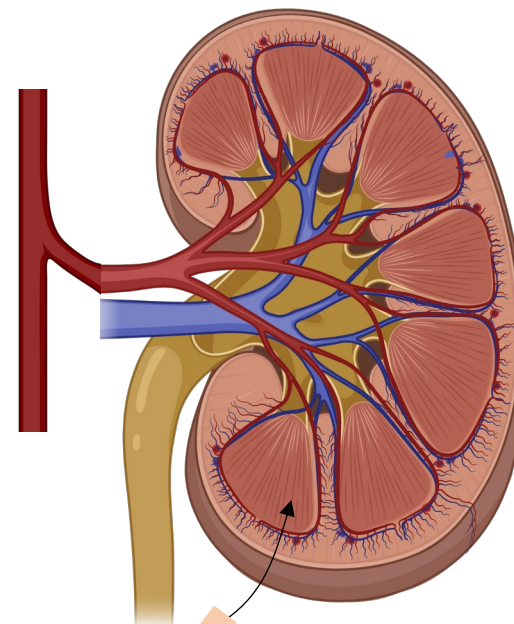
CASES

Calcium Channel Blocker (CCB)

Example	Dihydropyridines: Amlodipine 5-10mg PO daily
Best Candidate	Most patients. Well tolerated. Good option for late CKD.
Side Effects and Monitoring	Avoid in HFrEF, amlodipine may be used if necessary. Dose dependent pedal edema.

Four First Line Agents

Thiazide



Angiotensin II

ACE Inhibitor

Angiotensin Receptor Blocker (ARB)

Why is HTN important?

Treatment

Goals

1st line agents

Dihydro CCB

ACE-I

ARB

Thiazide

Secondary causes

CASES

Calcium Channel Blocker (CCB)

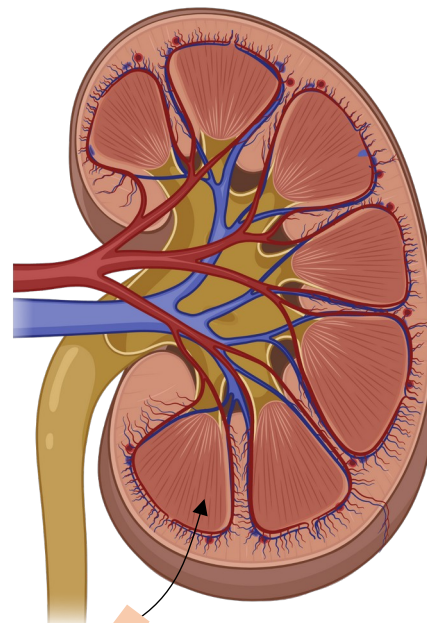
Thiazide

Example	Lisinopril 5-40mg PO daily
Best Candidate	Most patients. Well tolerated. Indications for CVD and proteinuria.
Side Effects and Monitoring	Do NOT use with ARB. Obtain BMP 1-2 weeks after initiation to monitor K and Cr. GFR ↓ of $\leq 30\%$ is expected and acceptable. Risk of angioedema.

ACE Inhibitor

Angiotensin II

Angiotensin Receptor Blocker (ARB)



Why is HTN important?

Treatment

Goals

1st line agents

Dihydro CCB

ACE-I

ARB

Thiazide

Secondary causes

CASES

Four First Line Agents

Calcium Channel Blocker (CCB)

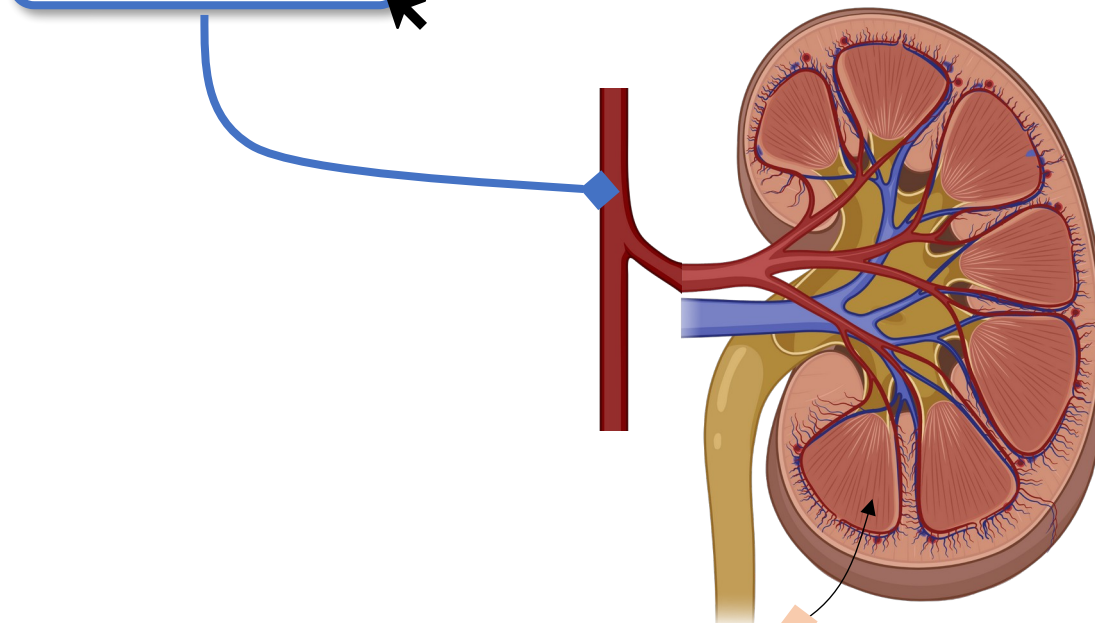
Thiazide

ACE Inhibitor

Angiotensin Receptor Blocker (ARB)

Angiotensin II

Example	Losartan 25-100mg po daily
Best Candidate	Did not tolerate ACE-I
Side Effects and Monitoring	Same as ACE-I Less risk of cough and angioedema



Why is HTN important?

Treatment

Goals

1st line agents

Dihydro CCB

ACE-I

ARB

Thiazide

Secondary causes

CASES

Four First Line Agents

Calcium Channel Blocker (CCB)

ACE Inhibitor

Angiotensin Receptor Blocker (ARB)

Thiazide

Example

Chlorthalidone
(preferred)
12.5-25mg PO daily

Best Candidate

Age <65 with venous insufficiency

Side Effects and Monitoring

Obtain BMP 2 weeks after initiation.
Monitor for low Na & low K.
Increased risk of gout.

Why is HTN important?

Treatment

Secondary causes

CASES

When to Suspect Secondary Cause

1. Severe or resistant hypertension
2. Acute rise or increased lability in BP
3. Before puberty OR less than 30 years without obesity or family history
4. Associated electrolyte abnormalities

Common Differential

Renovascular

Hyperaldosteronism

Sleep Apnea

Other Endocrine

Click each category button until cursor disappears

Clinical Signs

>50% ↓GFR with initiation of ACE I

Unexplained hypokalemia (sometimes)

Obesity, Snoring (+STOP BANG or + Epworth Sleepiness Scale)

Signs of **Cushing's**, **pheochromocytoma**, **hyper/hypothyroid**, & **oral contraceptives**

Diagnosis

Renal US w/ Doppler
CTA
MRA

Aldo/Renin > 20
(Off of spironolactone)
↓
CT Abd/Pelvis w/ IV contrast

Polysomnography

Salivary cortisol or dexamethasone suppression test,
TSH ± fT4,
Urine/serum, metanephrines

Why is HTN important?

Treatment

Secondary causes

CASES

1-2

3-4

Case 1

45 year old man presents to clinic with a single BP reading of 145/90.

What are the next steps to determine if this patient has HTN?

1. Recheck the measurement after sitting calmly for 5 min, arm relaxed, appropriate cuff size.
2. Confirm persistent elevation with 2+ readings in or out of the office

If diagnosed with HTN, what additional work-up is needed to further evaluate the cause?

1. Risk assessment with DM screening, lipids, Cr, UPC +/- ECG
2. No indication to evaluate for secondary cause.

What are the appropriate first-line agents & doses?

- Amlodipine 5-10mg qD
- Lisinopril 10-20mg po qD
- Losartan 25-50mg qD
- Chlorthalidone 12.5-25 mg qD

What lab surveillance, follow-up and titration would you recommend?

1. Home monitoring with evaluation of tolerance and response in 2-4 wks.
2. If starting ACE, ARB, or thiazide, recheck BMP in 1-2 weeks

Case 2

55 year old African-American woman returns to clinic for BP follow-up. She has no known heart or kidney disease and is a never smoker.

Three months ago, she was diagnosed with HTN with an average home reading of 155/95. She was started on amlodipine 10mg and has since had an average BP of 145/90, BMI 29. Normal A1c and lipid panel.

What is the next best step in her management?

Incorrect. No red flags for secondary cause of HTN.

Not 1st line. Only for refractory HTN, acute management or ↓ afterload.

Amlodipine is unlikely to have added benefit with BID dosing.

May be indicated in the future but current 10 y ASCVD risk is <10% (6.6%)

BP is not at goal with max dose CCB. Add another 1st line agent.

Why is HTN important?

Treatment

Secondary causes

CASES

1-2

3-4

Case 3

60 year old woman presents for routine follow-up. She has a history of obesity and pre-diabetes, consumes 1-2 alcoholic drinks a night, and does not smoke. Her BP is 150/90 for the second consecutive visit.

She prefers to avoid medications and wants to trial lifestyle measures alone.

What evidence-based lifestyle modifications to reduce BP?

Lifestyle change	BP effect
Dietary Na < 2 g/d	↓ SBP 5/DBP 2.5
High K diet (if no CKD or drug interaction)	↓ SBP 5/DBP 3
Lower alcohol intake	≥2 drinks/ day in F and ≥3 drinks/day in M significantly ↑ rates of HTN.
Weight loss	↓ 0.5-2 mmHg / every 1 kg weight loss
Exercise 3-4 sessions/ wk for ~ 40 min	↓ SBP 4-6/ DBP 3

Case 4

60 year old man with a history of gout and peripheral arterial disease presents with a BP 160/95. At his last few visits, his BP has been in a similar range. He does not check his BP at home.

What is your goal BP for this patient?

<130/80. High risk because of ASCVD (PAD).

What medications would you start?

Most patients that are >20/10 above BP goal will need 2 agents. You can add meds sequentially or start 2 meds simultaneously. One preferred combination is an ACEi + CCB because this demonstrated reduced CV risk compared to ACEi + thiazide (ACCOMPLISH trial)

If this patient told you that his BPs were normal at home, would you still start meds for HTN?

White coat hypertension is associated with increased risk of cardiovascular events and all cause mortality.

Most experts still **recommend against starting medications** empirically at this point, but rather aggressively managing other cardiovascular risk factors.

1. Benefits of treatment

Example: 50-year-old woman who smokes tobacco, otherwise healthy with sustained BP >150/90

→ If treated to BP <140/85

CV Events Over 10 years
Absolute risk reduction = 10%
NNT = 10

Mortality Over 10 years
Absolute risk reduction = 6%
NNT = 16

2. Goal blood pressure for everyone is:

SBP <120 / DBP <80

...but, pharmacologic treatment targets less aggressive:

High risk patients?

SBP <130 / DBP <80

Patient without high-risk diagnosis?

SBP <140 / DBP <90

Calcium Channel Blocker (CCB)

Example	Amlodipine 5-10mg PO daily
Best Candidate	Most patients. Well tolerated. Good option for late CKD.
Side Effects and Monitoring	Avoid in HFrEF, amlodipine may be used if necessary. Dose dependent pedal edema.

Example	Lisinopril 5-40mg PO daily
Best Candidate	Most patients. Well tolerated. Indications for CVD and proteinuria.
Side Effects and Monitoring	Obtain BMP 1-2 weeks after initiation to monitor K and Cr. Acceptable GFR reduction of $\leq 30\%$. Risk of angioedema.

ACE Inhibitor

3. Four First Line Agents

Thiazide

Example	Chlorthalidone (preferred) 12.5-25mg PO daily
Best Candidate	Age <65 with venous insufficiency
Side Effects and Monitoring	Obtain BMP 2 weeks after initiation. Monitor for low Na & low K. Increased risk of gout. Not effective with GFR <30

Example	Losartan 25-100mg po daily
Best Candidate	Did not tolerate ACE-I
Side Effects and Monitoring	Same as ACE-I, less risk of cough and angioedema

Angiotensin Receptor Blocker (ARB)

Angiotensin II

4. When to Suspect Secondary Cause

1. Severe or resistant hypertension
2. Acute rise or increased lability in BP

3. Before puberty OR less than 30 years without obesity or family history
4. Associated electrolyte abnormalities

Common Differential

Renovascular

Hyperaldosteronism

Sleep Apnea

Other Endocrine

2025 AHA/ACC Guideline Update

- **Uniform BP goal <130/80 for MOST adults — replaces the split low-risk <140/90 vs high-risk <130/80 shown on the board. Confirm dx with out-of-office (home/ABPM) readings.**
- Risk tool changed: PREVENT equation replaces the Pooled Cohort Equations for 10-yr CVD risk (ages 30–79; includes eGFR, statin use, social factors).
- Stage 1 (130–139/80–89): if known CVD/DM/CKD or 10-yr risk $\geq 7.5\%$ → start a drug; if $< 7.5\%$ → lifestyle 3–6 mo first, then a drug if still $\geq 130/80$.
- Stage 2 ($\geq 140/90$): start TWO first-line agents, ideally as a single-pill combination. Four first-line classes unchanged (thiazide-like, dihydropyridine CCB, ACEi, ARB).
- Resistant HTN (uncontrolled on ACEi/ARB + CCB + thiazide-like): add an MRA (spironolactone) if eGFR ≥ 45 ; apocitentan (endothelin antagonist) is a newer add-on. Obesity: GLP-1 receptor agonists now endorsed for weight-driven BP reduction.

Source: 2025 AHA/ACC Hypertension Guideline (JACC / Hypertension 2025).

Take-Home Points

- **Treat to <130/80 for most adults (2025 AHA/ACC); the board's split target reflects the older 2017 risk-based framing.**
- Decide WHO to start on a drug by absolute risk: established CVD/DM/CKD or 10-yr risk $\geq 7.5\%$ (PREVENT) → treat at $\geq 130/80$; lower risk → lifestyle first.
- Four first-line agents: thiazide-like diuretic (chlorthalidone preferred), dihydropyridine CCB (amlodipine), ACEi (lisinopril), ARB (losartan). Never combine ACEi + ARB. Stage 2 → start two, ideally one pill.
- Lifestyle works and is foundational: DASH diet, sodium <2 g/d, high-potassium diet, weight loss, exercise, limit alcohol — each worth a few mmHg.
- Suspect a secondary cause if HTN is severe/resistant, of acute onset or labile, age <30 without risk factors, or has unexplained electrolyte abnormalities — hyperaldosteronism is the most common (low threshold for an aldosterone:renin ratio).

Attribution & Optimization Notes

- **Original: "Management of Chronic Hypertension," TeachIM.org — Fainstad B, Guinn K, Melton M, Trefilek B; ed. Brett M. Published Sept 2021, reviewed Oct 2022.**
- Source URL: teachim.org/teaching_material/htn/
- License: Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0). Educational use only; not medical advice.
- Corrections: none required — no factual or numeric errors found in the deck on review.
- Added context: 2025 AHA/ACC guideline update slide (uniform <130/80 goal, PREVENT calculator, stage-2 two-drug start, MRA/aprocitentan for resistant HTN, GLP-1 for obesity). Original teaching slides preserved unchanged; content appended only.
- All 12 original animated slides and graphics preserved. Optimized 2026-06-10. Clinical content should be clinician-reviewed before teaching.